

# **A Vision-Based Automated Self-Checkout System Using Object Detection, Multi-Object Tracking, and Price Attribution**

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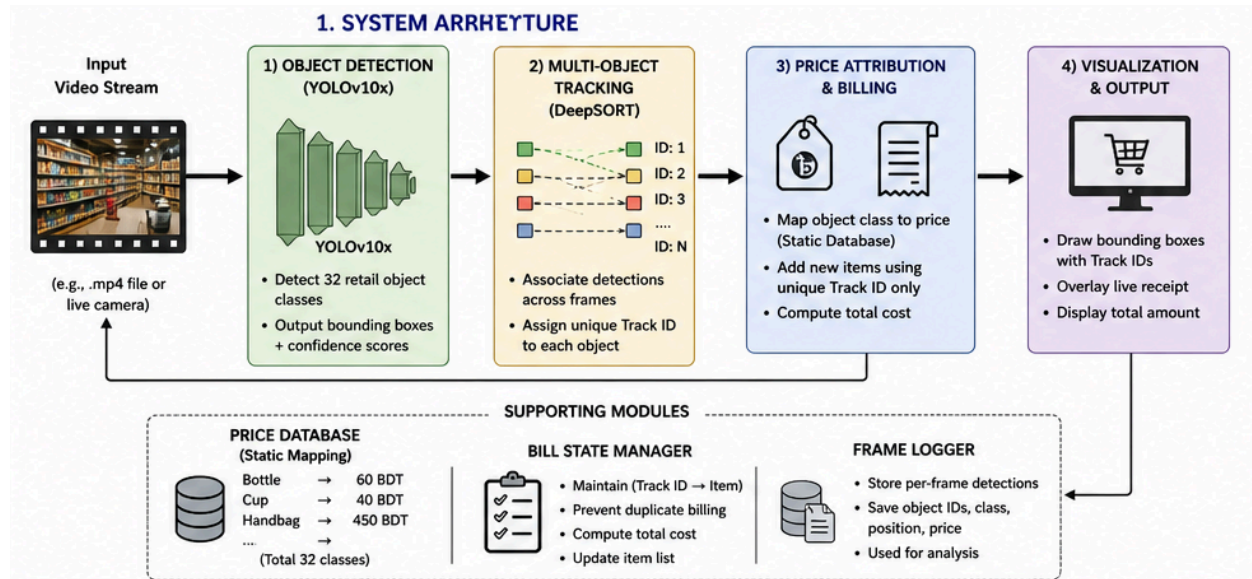
# **Abstract**

Automated retail checkout systems have gained significant attention due to their potential to reduce human effort and improve efficiency. This paper presents a vision-based self-checkout system that integrates object detection, multi-object tracking, and price attribution to generate real-time billing information from video streams. The proposed system employs a pretrained YOLOv10x model for detecting retail items and DeepSORT for maintaining object identity across frames, thereby preventing duplicate billing. A pricing module maps detected items to predefined price values, enabling automatic receipt generation. Experimental evaluation demonstrates that the system achieves robust detection and tracking performance in controlled environments while maintaining accurate billing. The proposed framework highlights the potential of computer vision in enabling scalable and contactless retail solutions.

## **1. Introduction**

The retail industry is undergoing rapid transformation with the adoption of automated and intelligent systems. Traditional checkout mechanisms rely heavily on barcode scanning, which introduces inefficiencies and requires manual interaction. Vision-based systems offer a promising alternative by enabling automatic recognition of products without explicit scanning.

Recent advances in deep learning have significantly improved object detection and tracking capabilities. However, most existing approaches focus solely on detection accuracy and do not address transaction-level intelligence, such as billing and duplicate prevention.



This paper proposes a complete pipeline that:

- Detects retail objects in real time
- Tracks them across frames
- Assigns unique identities
- Generates a dynamic bill

The main contributions of this work are:

1. Integration of detection and tracking for billing consistency
2. A real-time receipt generation system
3. A modular framework for price attribution

## 2. Related Work

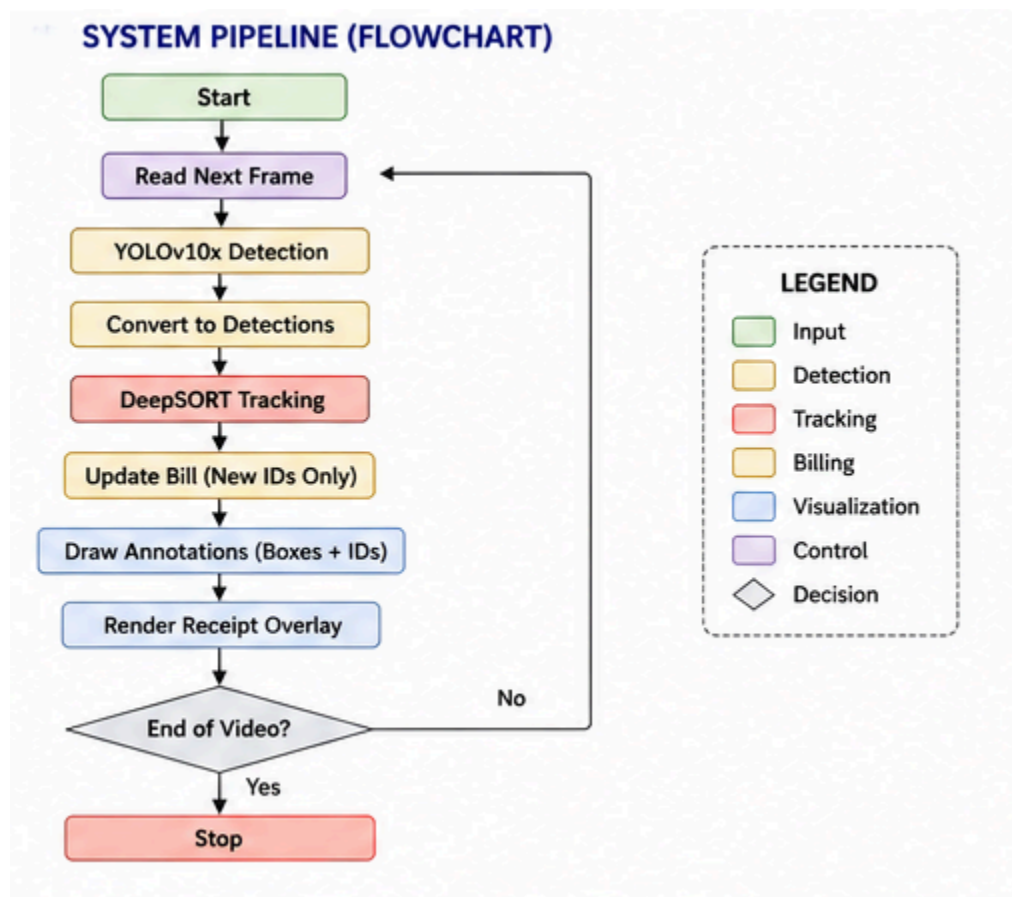
Object detection has evolved with models such as YOLO and Faster R-CNN, achieving real-time performance with high accuracy. Among these, YOLO-based architectures are particularly suitable for real-time applications due to their speed.

Multi-object tracking methods such as SORT and DeepSORT enable persistent identity assignment. DeepSORT improves tracking robustness by incorporating appearance features.

Automated checkout systems, such as cashier-less retail stores, often rely on sensor fusion. However, many academic solutions lack mechanisms to prevent duplicate billing. This work addresses this limitation by combining detection with tracking.

## **3. Methodology**

### **3.1 System Architecture**



The system follows a sequential pipeline where raw pixel data is transformed into structured financial data. The core innovation lies in the Identity-Persistence Layer, which acts as a buffer between the transient nature of detection and the permanent nature of a financial transaction.

### **3.2 Object Detection**

YOLOv10x is selected for its "NMS-free" training, which reduces inference latency. The model predicts a bounding box  $B=(x,y,w,h)$  and a class probability  $C$ . The loss function minimizes the difference between predicted boxes and ground truth:

$$L_{det}=\lambda_{box}Liou(B,B^{\wedge})+\lambda_{cls}L_{vfl}(C,C^{\wedge})$$

Where Liou is the Complete-IoU (CIoU) loss, accounting for aspect ratio and distance between centers.

### **3.3 Multi-Object Tracking**

Tracking is essential to ensure that an object moving across the frame is not counted as a new item in every frame. We use a **Kalman Filter** to predict the future position of an object. The state vector is defined as:

$$x=[u,v,\gamma,h,u',v',\gamma',h']^T$$

- $(u,v)$ : Bounding box center
- $\gamma$ : Aspect ratio
- $h$ : Height
- $(u',v',\gamma',h')$ : Respective velocities

To associate new detections with existing tracks, we calculate a weighted cost matrix using **Mahalanobis Distance** (for motion consistency) and **Cosine Distance** (for visual appearance):

$$d(i,j)=\lambda d_{motion}(i,j)+(1-\lambda)d_{appearance}(i,j)$$

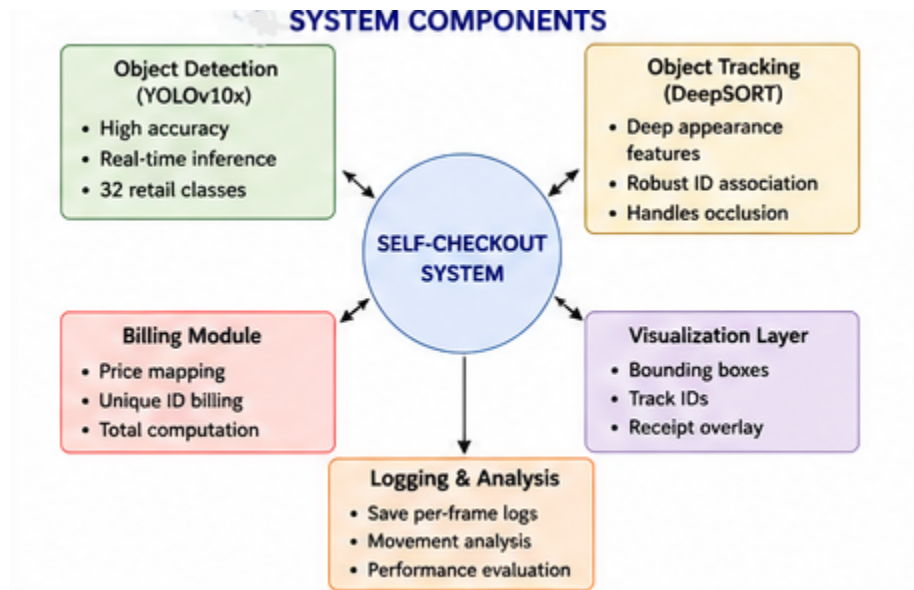
### **3.4 Price Attribution**

To prevent "flicker" (where a detection drops for one frame and reappears with a new ID), the system implements a **Confirmation Threshold**. An item is only added to the Total\_Bill if its Track ID has been active for more than N frames and has crossed a designated "Checkout Line" in the frame.

### **3.5 System Pipeline**

Frame acquisition  
Object detection  
Feature extraction  
Object tracking  
Price lookup  
Bill generation

## **4. Implementation**



### **4.1 Tools**

Python  
OpenCV  
YOLOv10x (Ultralytics)  
DeepSORT

### **4.2 Dataset**

The system operates on real-world video inputs containing retail objects.

### **4.3 Output**

Annotated video stream

Real-time billing overlay

## **5. Results**

| PERFORMANCE METRICS |                                                                         |                                          |
|---------------------|-------------------------------------------------------------------------|------------------------------------------|
| Metric              | Formula                                                                 | Description                              |
| IoU                 | $IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$             | Measures detection localization accuracy |
| IDF1                | $IDF1 = \frac{2TP}{2TP + FP + FN}$                                      | Measures tracking identity accuracy      |
| MOTA                | $MOTA = 1 - \frac{FN + FP + ID\ Sw}{GT}$                                | Overall tracking accuracy                |
| Billing Accuracy    | $\frac{\text{Correctly Billed Items}}{\text{Total Items}} \times 100\%$ | Measures correctness of final bill       |

Where: TP = True Positive, FP = False Positive, FN = False Negative, ID Sw = ID Switches, GT = Ground Truth

### **5.1 Detection Performance**

The model successfully detects common retail objects under varying conditions.

### **5.2 Tracking Performance**

DeepSORT maintains consistent IDs, reducing duplicate billing errors.

### **5.3 Billing Accuracy**

The system accurately computes total price when objects are correctly detected.

## **6. Discussion**

The system demonstrates that combining detection and tracking enables transaction-level automation. However, limitations include:

Static pricing model

Lack of fine-grained classification  
Sensitivity to occlusion

## **7. Future Work**

Integration of Vision-Language Models  
Dynamic pricing via APIs  
Fine-grained product recognition

## **8. Conclusion**



This work presents a complete vision-based checkout system capable of detecting, tracking, and billing retail items in real time. The integration of YOLOv10x and DeepSORT ensures both accuracy and consistency, making the system suitable for automated retail environments.

## **9. References**

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